BASE CHANGE SHIFTS THE DELTA-DIMENSION OF EVERY IRREDUCIBLE COMPONENT BY THE CONSTANT C

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVH (sources/stacks-chow/pinned-original-source.tex line 15139).

Allowed standard prerequisites (editorial).

- Flatness of base change and lifting of generalizations along flat morphisms. Morphisms, Lemmas labelled morphisms-lemma-base-change-flat and morphisms-lemma-generalizations-lift-flat, cited at proof lines 15146-15153. Shows X' -> X is flat and that generic points of components of X' lie over the generic point of X.
- Fibres and their topological description. Schemes, Lemma labelled schemes-lemma-fibre-topological, cited at proof lines 15157-15160. Identifies S'_s with g^{-1}(s) so that x' can be seen as a point of S'_s x_s x.
- Equality of fibre dimensions and of transcendence degrees after base change. Morphisms, Lemmas labelled morphisms-lemma-dimension-fibre-after-base-change and morphisms-lemma-dimension-fibre-at-a-point, cited at proof lines 15169-15175. Gives dim_x(X_s) = dim_{x'}(X_{s'}) and the equality of the two transcendence degrees.
- The delta-dimension via the dimension function and the transcendence degree. Same chapter, Definition labelled definition-delta-dimension, cited at proof lines 15177-15179. Converts the displayed equality into the statement about dim_{delta'}(Z').

Source reference. Original reference 0FVG: sources/stacks-chow/pinned-original-source.tex, source lines 15123–15123. Same-chapter pointer into the author environment situation at source line 15123; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 15148 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15152 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01K1: sources/stacks-chow/tags.txt, source lines 2034–2034. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15156 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15166 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15172 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02QP: sources/stacks-chow/pinned-original-source.tex, source lines 1504–1504. Same-chapter pointer into the author environment definition at source line 1504; invoked at proof line 15178 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0FVG let $X \to S$ be locally of finite type. Denote $X' \to S'$ the base change by $S' \to S$. If X is integral with $\dim_\delta(X) = k$, then every irreducible component Z' of X' has $\dim_{\delta'}(Z') = k + c$,*

Proof. The projection $X' \to X$ is flat as a base change of the flat morphism $S' \to S$ (Morphisms, Lemma 01U9). Hence every generic point x' of an irreducible component of X' maps to the generic point $x \in X$ (because generalizations lift along $X' \to X$ by Morphisms, Lemma 03HV). Let $s \in S$ be the image of x. Recall that the scheme $S'_s = S' \times_S s$ has the same underlying topological space as $g^{-1}(\{s\})$ (Schemes, Lemma 01K1). We may view x' as a point of the scheme $S'_s \times_s x$ which comes equipped with a monomorphism $S'_s \times_s x \to S' \times_S X$. Of course, x' is a generic point of an irreducible component of $S'_s \times_s x$ as well. Using the flatness of $\mathrm{Spec}(\kappa(x)) \to \mathrm{Spec}(\kappa(s)) = s$ and arguing as above, we see that x' maps to a generic point s' of an irreducible component of $g^{-1}(\{s\})$. Hence $\delta'(s') = \delta(s) + c$ by assumption. We have $\dim_x(X_s) = \dim_{x'}(X_{s'})$ by Morphisms, Lemma 02FY. Since x is a generic point of an irreducible component X_s (this is an irreducible scheme but we don't need this) and x' is a generic point of an irreducible component of $X'_{s'}$ we conclude that $\mathrm{trdeg}_{\kappa(s)}(\kappa(x)) = \mathrm{trdeg}_{\kappa(s')}(\kappa(x'))$ by Morphisms, Lemma 02FX. Then

$$\delta_{X'/S'}(x') = \delta(s') + \mathrm{trdeg}_{\kappa(s')}(\kappa(x')) = \delta(s) + c + \mathrm{trdeg}_{\kappa(s)}(\kappa(x)) = \delta_{X/S}(x) + c$$

This proves what we want by Definition 02QP. □